

ARRAYS

Q4) Sort Array In Waveform Order

TCS Interview Question

Question description

Ravi participated in TCS off campus interview at reputed institution, one of the technical question he has to complete with in the given time, where you need to sort the array in the waveform. There might be multiple possible output of the program, the following pattern output is appreciated.

Function Description

This is a simple method of solving this question which contains basic 2 steps and they are as follow

Step : 1 – Sort the array in ascending order.

Step : 2 – Swap all adjacent elements of the array

Let us consider the input array be [3, 6, 5, 10, 7, 20]. After sorting, we get [3, 5, 6, 7, 10, 20]. After swapping adjacent elements, we get [5, 3, 7, 6, 20, 10].

CODE:

```
C main.c
1  #include <stdio.h>
2
3  void sort(int arr[], int n) {
4      for (int i = 0; i < n - 1; i++) {
5          for (int j = 0; j < n - i - 1; j++) {
6              if (arr[j] > arr[j + 1]) {
7                  int temp = arr[j];
8                  arr[j] = arr[j + 1];
9                  arr[j + 1] = temp;
10             }
11         }
12     }
13 }
14
15 int main(void) {
16     int n;
17     if (scanf("%d", &n) != 1) {
18         return 0;
19     }
20
21     int arr[100];
22     for (int i = 0; i < n; i++) {
23         scanf("%d", &arr[i]);
24     }
25
26     sort(arr, n);
27
28     for (int i = 0; i < n - 1; i += 2) {
29         int temp = arr[i];
30         arr[i] = arr[i + 1];
31         arr[i + 1] = temp;
32     }
33
34     for (int i = 0; i < n; i++) {
35         printf("%d ", arr[i]);
36     }
37     printf("\n");
38
39     return 0;
40 }
```

OUTPUT:

```
PS C:\Users\CH.SC.U4CSE25255\Projects\test> ./main
6
3 6 5 10 7 20
5 3 7 6 20 10
```